

No.	even or odd program
1	Print Hello World
2	Swap Two variables
3	Even or odd:- check number
4	Even or odd:- without any condition?
5	Even or odd:- Print number

Ques. Print Hello World?

```
print('Hello World!!!')
```

Output:- Hello World!!!

Swap Two variables

```
a = 11
b = 7

# -- Using a Temporary variable
temp = a
a = b
b = temp

print(a) # 7
print(b) # 11

# -- Without a temporary variable
a, b = b, a

print(a) # 7
print(b) # 11

# -- Using arithmetic operators
a = a + b # a = 18, b = 7
b = a - b # a = 18, b = 11
a = a - b # a = 7, b = 11

print(a) # 7
print(b) # 11

# -- Using multiplication and division operator
# To Swap the values of two variables using Addition and subtraction operator
P = P * Q
Q = P / Q
P = P / Q

print ("The Value of P after swapping: ", P)
print ("The Value of Q after swapping: ", Q)

# -- XOR swap
P = P ^ Q
Q = P ^ Q
P = P ^ Q

print ("The Value of P after swapping: ", P)
print ("The Value of Q after swapping: ", Q)
```

Even Or Odd

Ques. Program to check if a number is Even or odd?

```
# 1 Option
num = int(input("Enter a number: "))
if (num % 2) == 0:
    print("{0} is Even".format(num))
else:
    print("{0} is Odd".format(num))

# 2 option using function
def evenOrOdd(n):
    if n%2 == 0:
        print('even Number hai')
    else:
        print('Odd Number hai')
evenOrOdd(6)
```

Check even or odd without any condition?

```
# Type 1
number = 61
list = ["even", "odd"]
print(list[number%2])

# Type 2
def check_even_odd(n):
    return ("Even" * (n & 1 == 0)) or ("Odd")
    # return ("Even" * (n %2==0)) or ("Odd")

# Test the function
number = 5
result = check_even_odd(number)
print(f"The number {number} is {result}.") # Output:- The number 5 is Odd.
```

Print the even number and odd number

```
num = int(input("Enter a number: "))
even = []
odd = []
for i in range(2,num+1):
    if(i%2==0):
        even.append(i)
    else:
        odd.append(i)

print(even) # Output:- [2, 4, 6, 8, 10, 12, 14]
print(odd)  # Output:- [3, 5, 7, 9, 11, 13, 15]
```